Laboratory Work #3

SQL Data Manipulation in PostgreSQL

Report

Database: DB26_Labs

DBMS: PostgreSQL 18

Environment: pgAdmin 4

Student: Nikita Gordeev

The report contains SQL execution evidence and verification screenshots.

Laboratory Work #3 - Overview

Objective

The purpose of this laboratory work is to practice basic data manipulation operations in PostgreSQL. The work demonstrates inserting sample data, updating existing records, deleting records according to conditions, and verifying the results with SELECT queries.

Operations demonstrated

- INSERT - population of tables with logically related sample records.
- UPDATE - modification of airline, airport, booking and baggage-check data.
- DELETE - removal of records that match date, price and status conditions.
- SELECT - verification of the database state before and after changes.
- Use of CURRENT_TIMESTAMP and conditional expressions in data preparation.

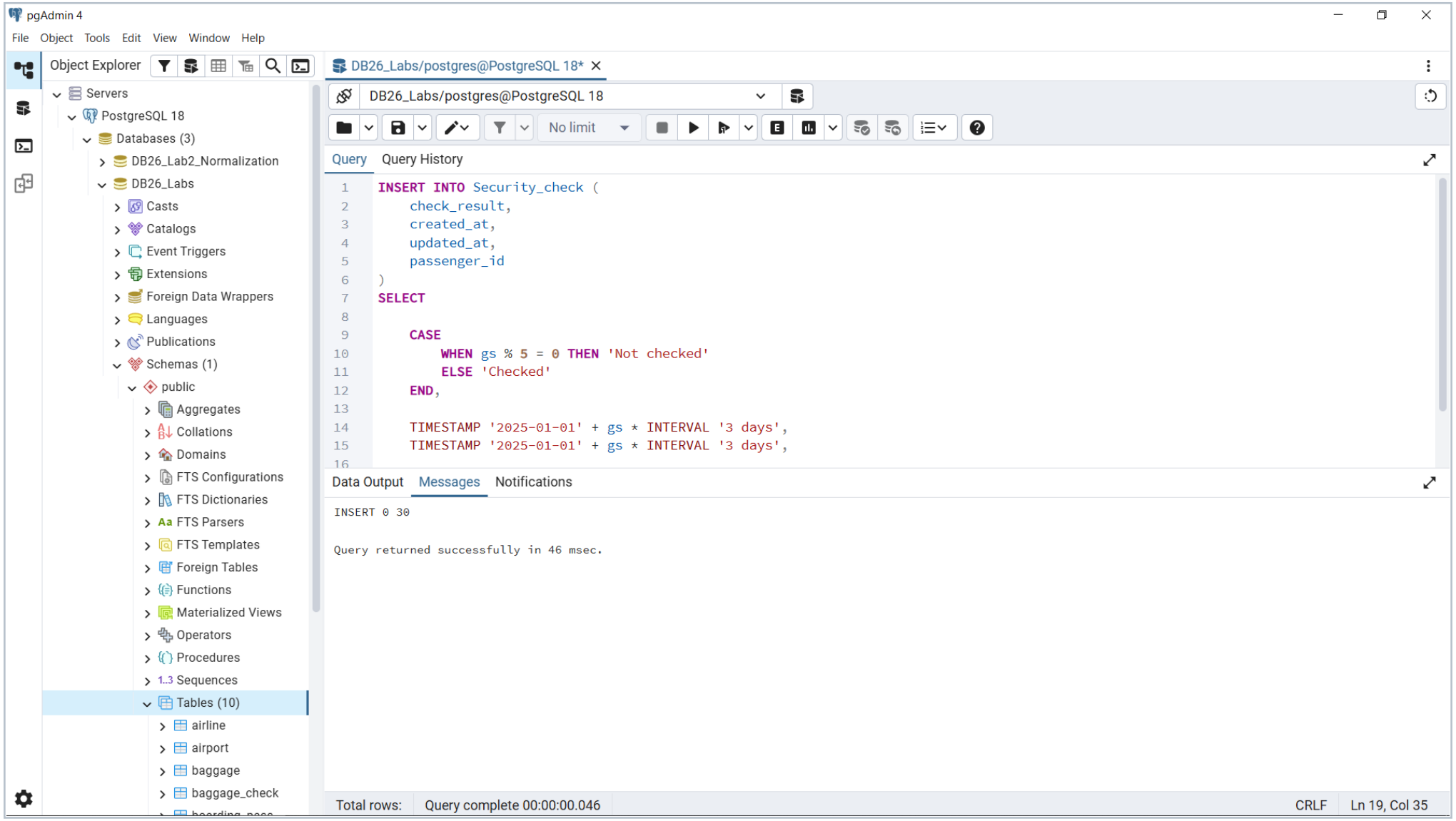
Evidence

The following pages contain 26 pgAdmin screenshots documenting successful execution and verification.

Execution Evidence 1

Screenshot_21(1).png

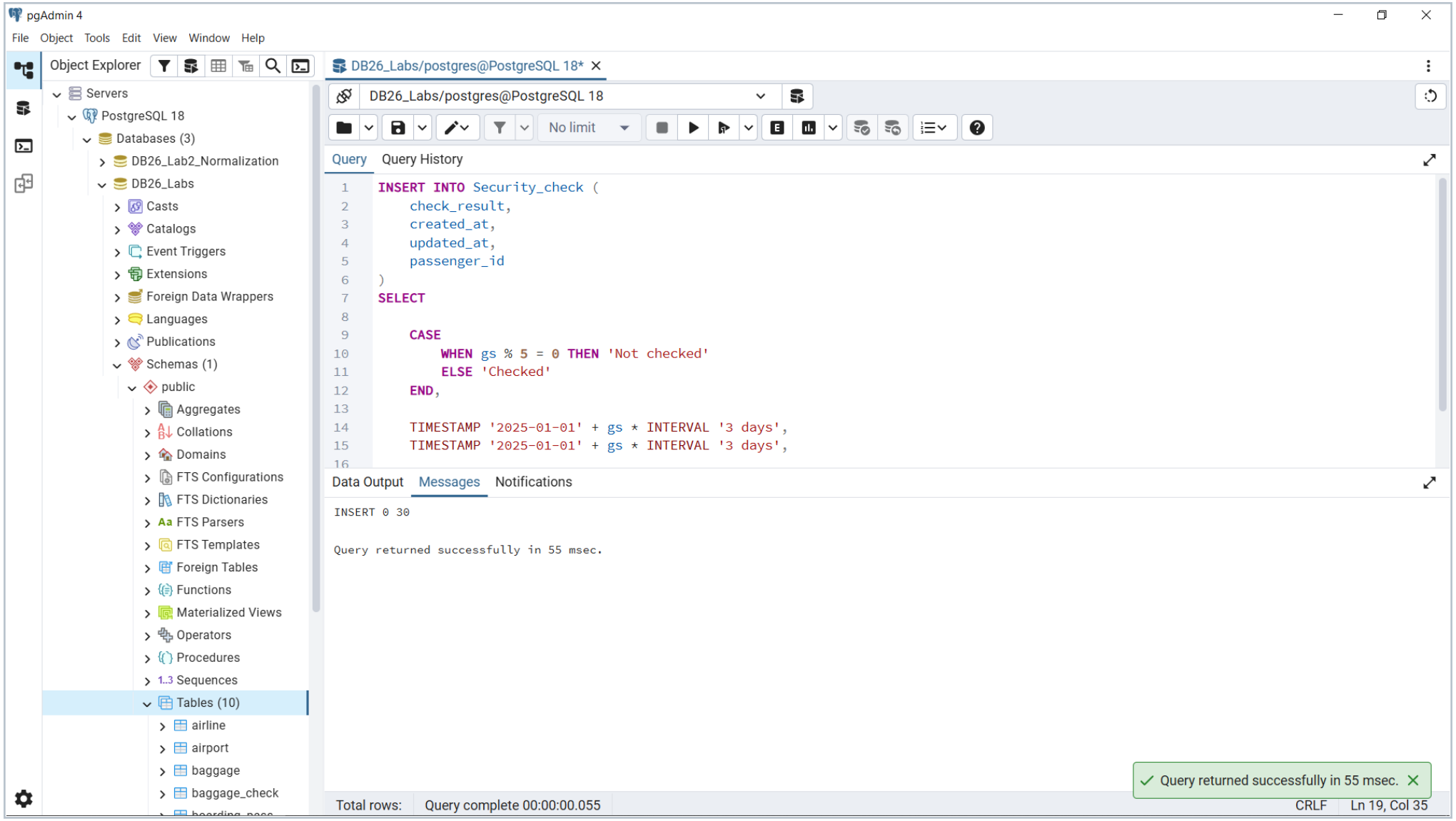
Description: Population and manipulation of related airport database tables.



Execution Evidence 2

Screenshot_22(1).png

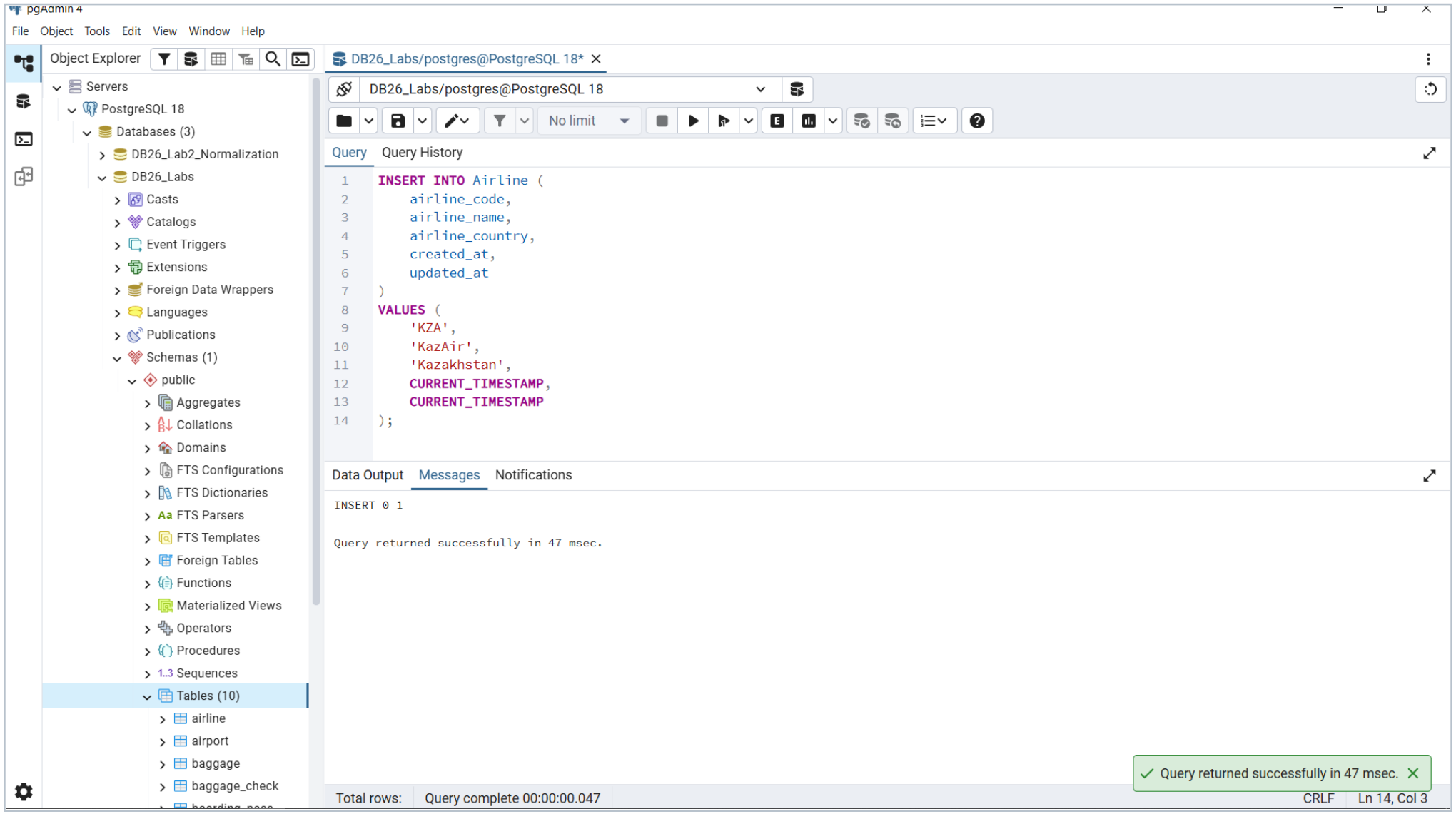
Description: Execution of data manipulation statements with successful insertion.



Execution Evidence 3

Screenshot_23(1).png

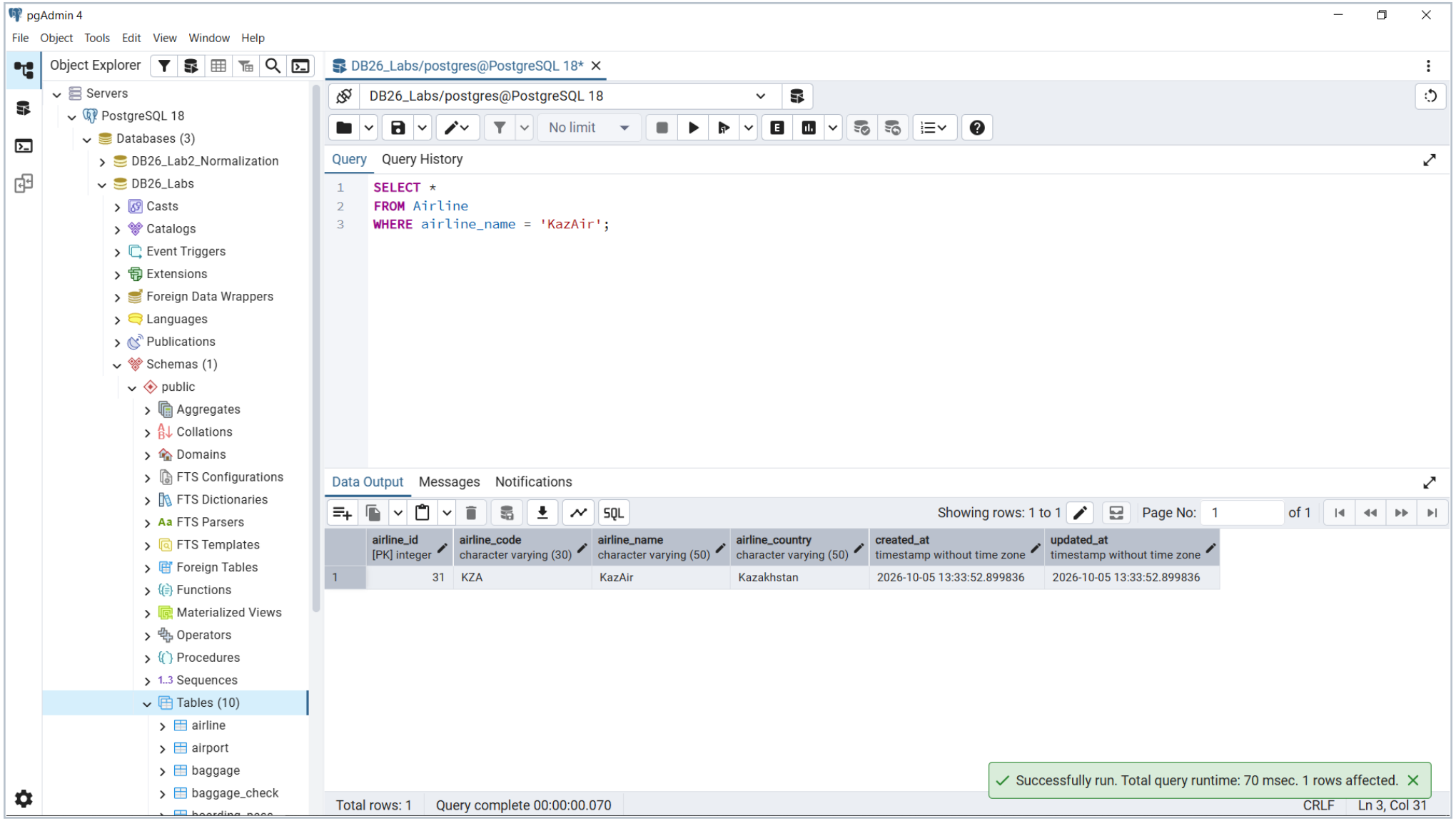
Description: Insertion of sample airline records and verification of results.



Execution Evidence 4

Screenshot_24.png

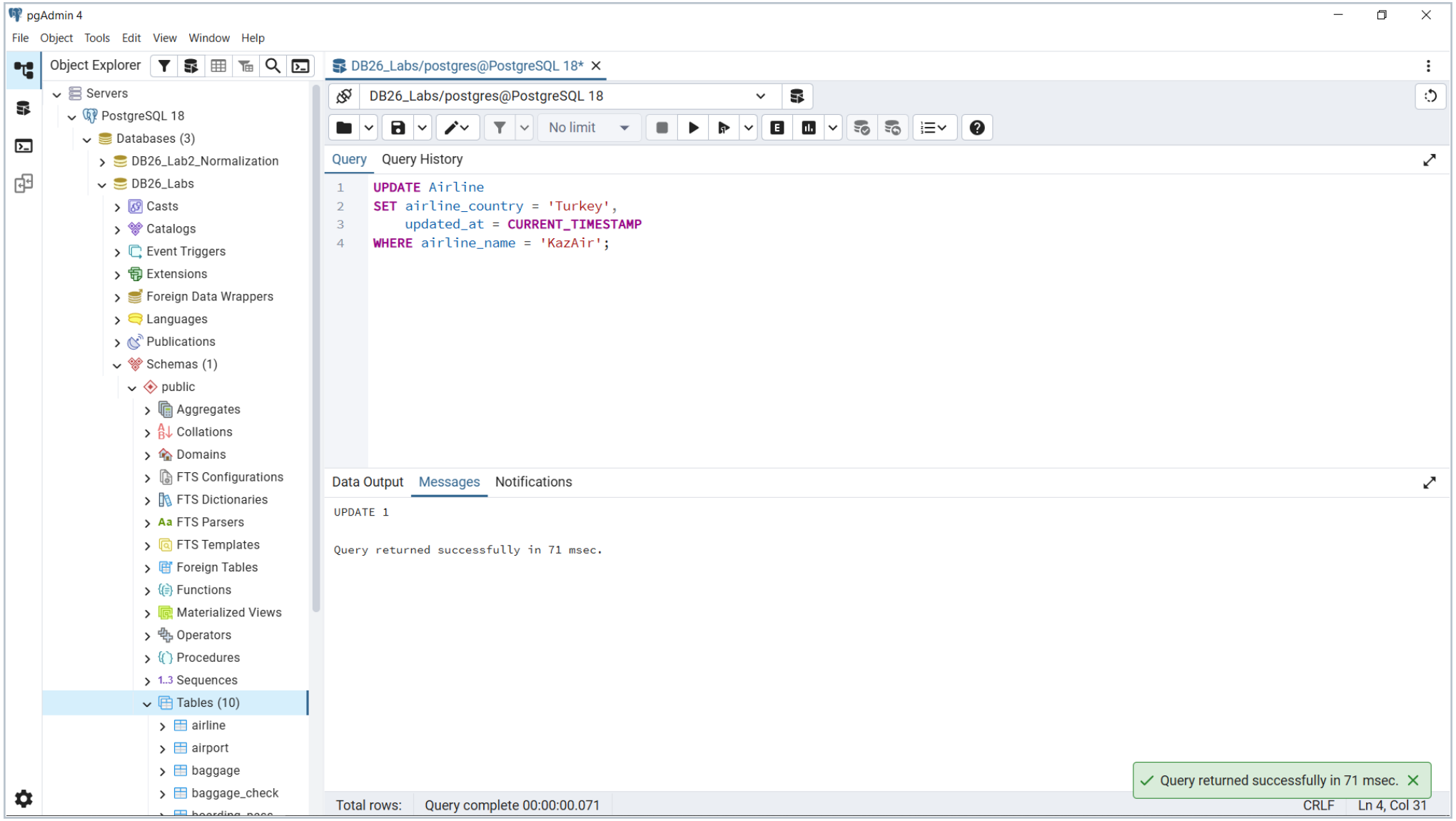
Description: Verification query showing inserted airline data.



Execution Evidence 5

Screenshot_25.png

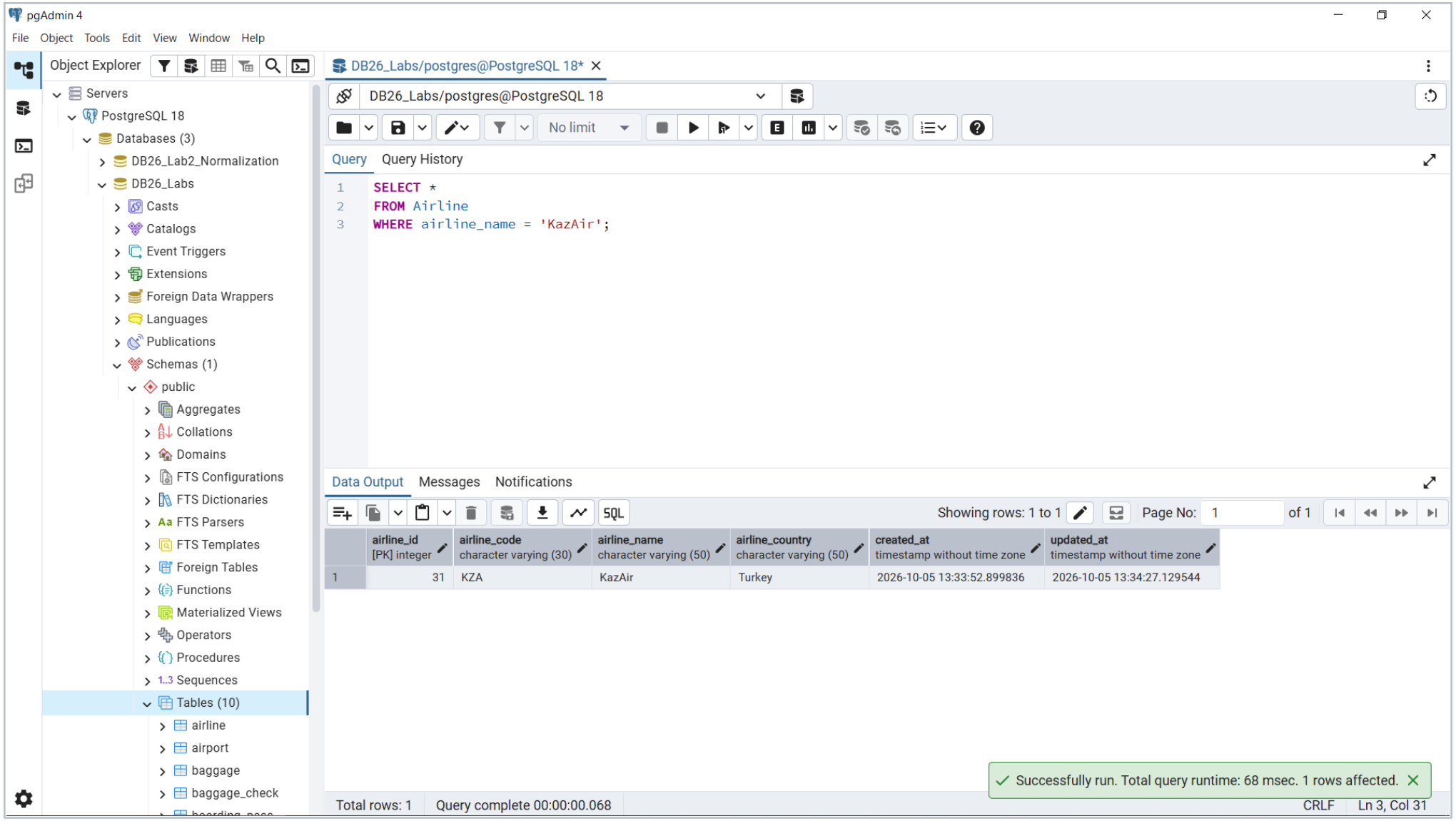
Description: Additional INSERT operation used to prepare the database for Lab 3 tasks.



Execution Evidence 6

Screenshot_26.png

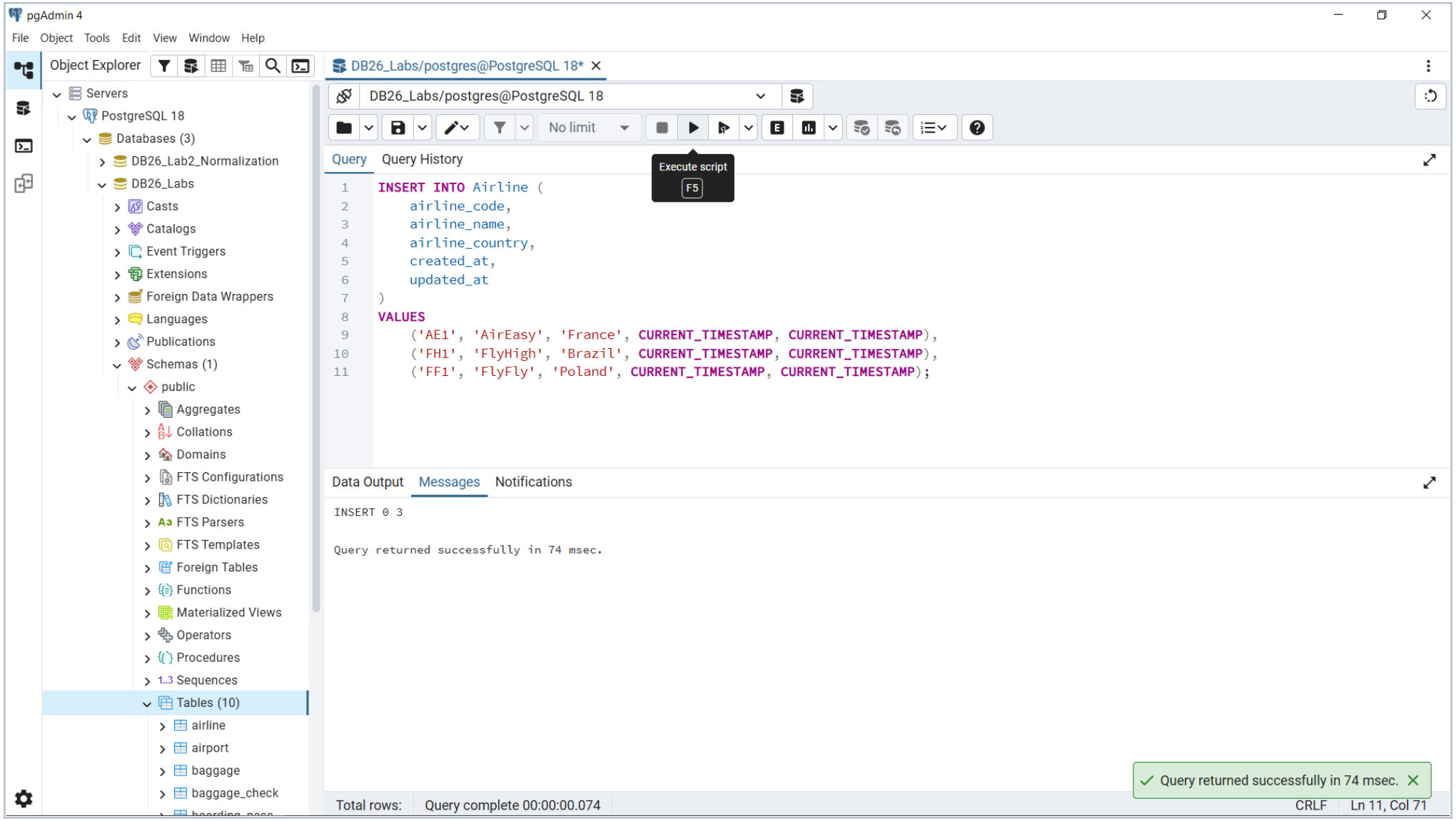
Description: Verification of inserted data in the target table.



Execution Evidence 7

Screenshot_27.png

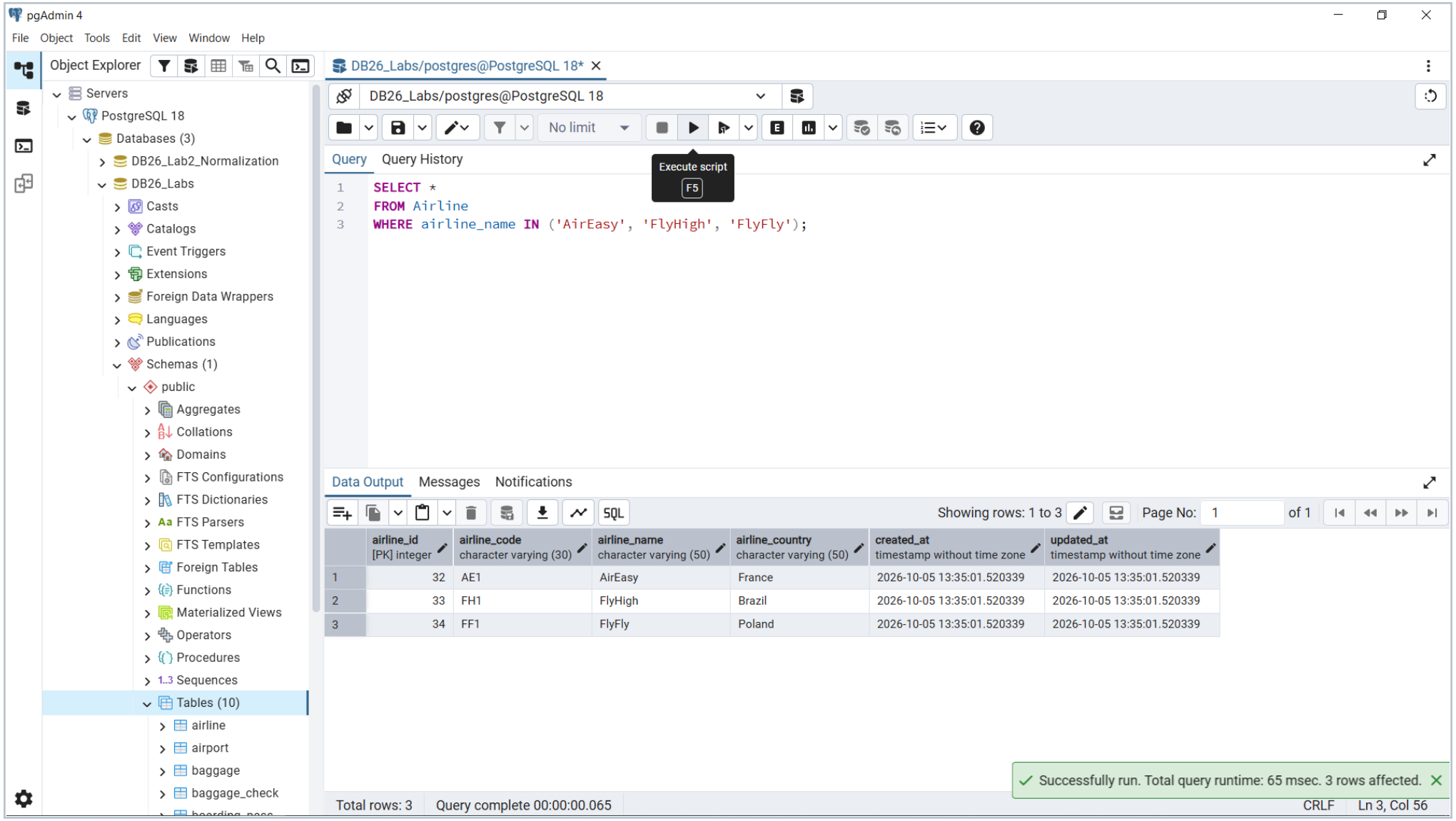
Description: Execution of a data modification statement in pgAdmin.



Execution Evidence 8

Screenshot_28.png

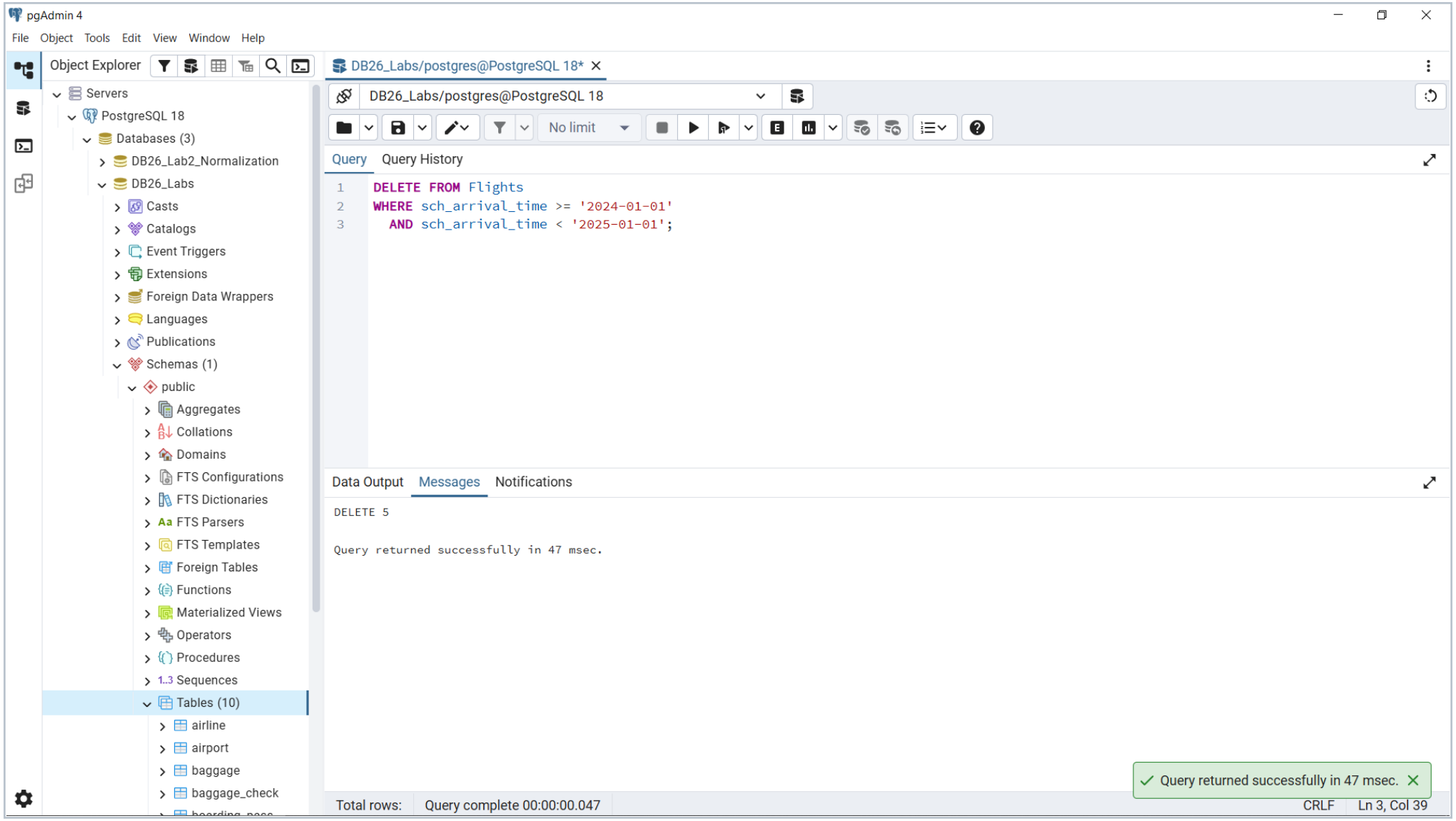
Description: Verification of the modified records.



Execution Evidence 9

Screenshot_29.png

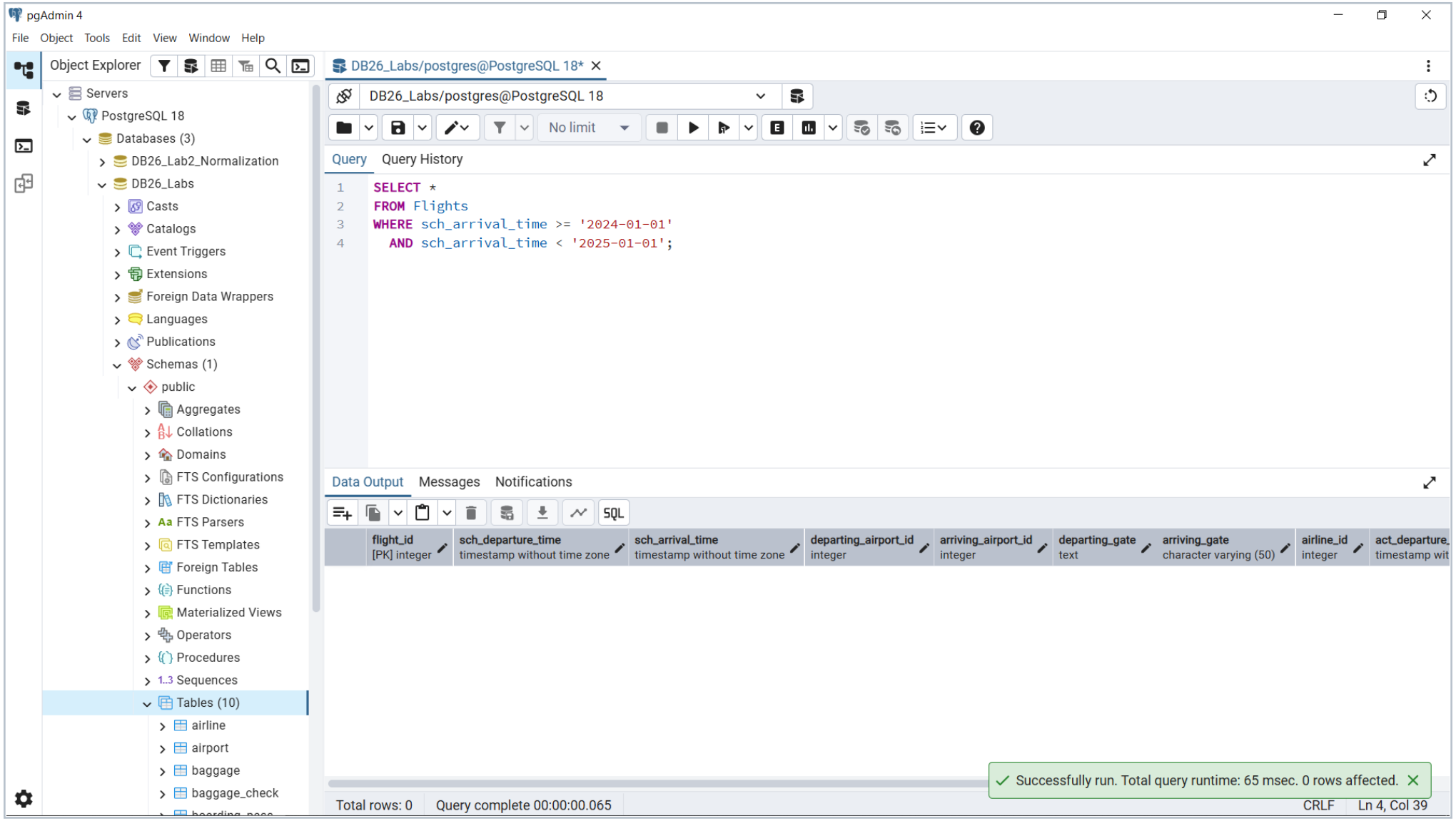
Description: DELETE operation for flights in the specified date range.



Execution Evidence 10

Screenshot_30.png

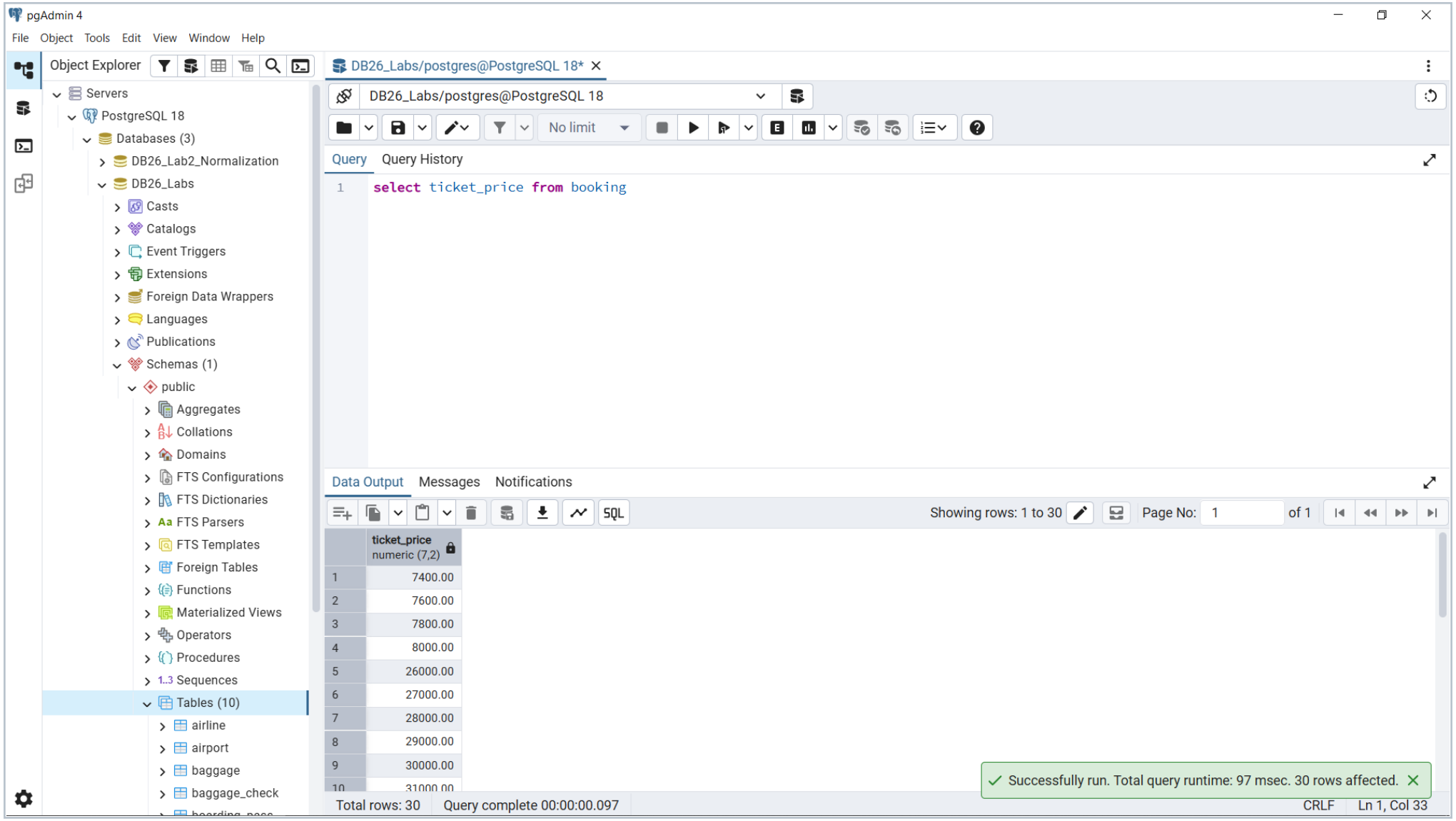
Description: Verification query after deleting flights from the selected date range.



Execution Evidence 11

Screenshot_31.png

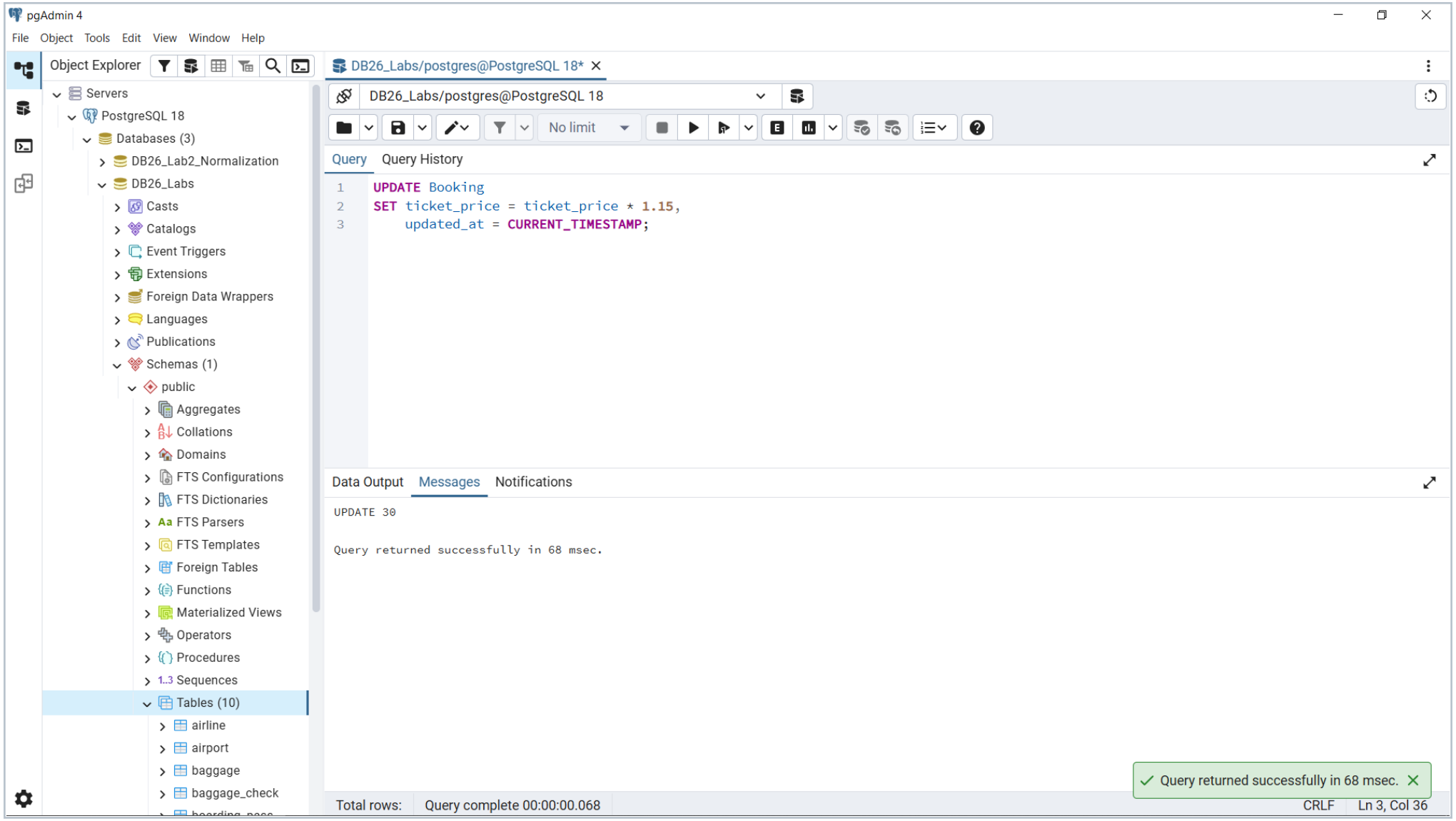
Description: Ticket price values before modification.



Execution Evidence 12

Screenshot_32.png

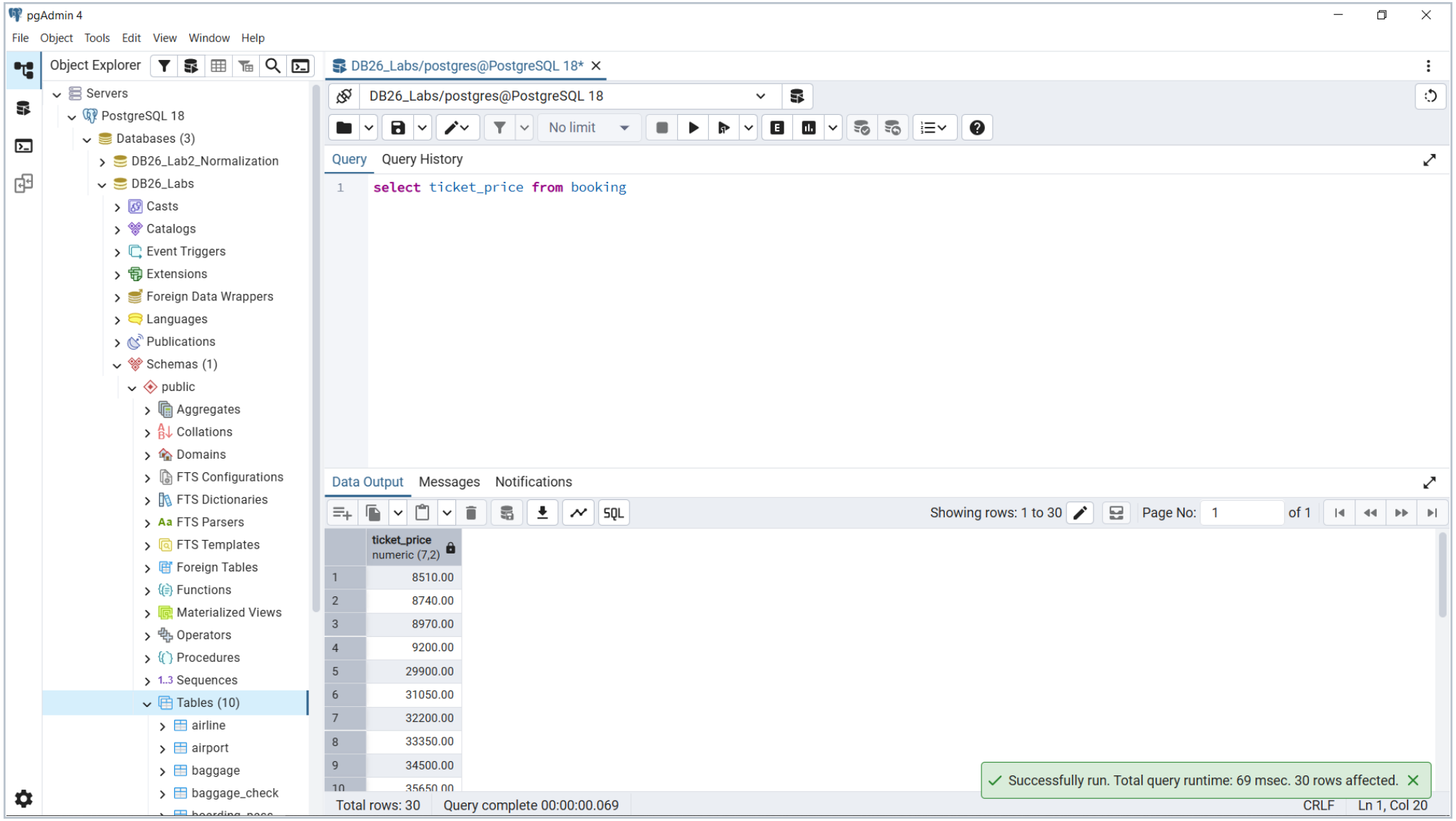
Description: UPDATE operation increasing booking ticket prices by 15 percent.



Execution Evidence 13

Screenshot_33.png

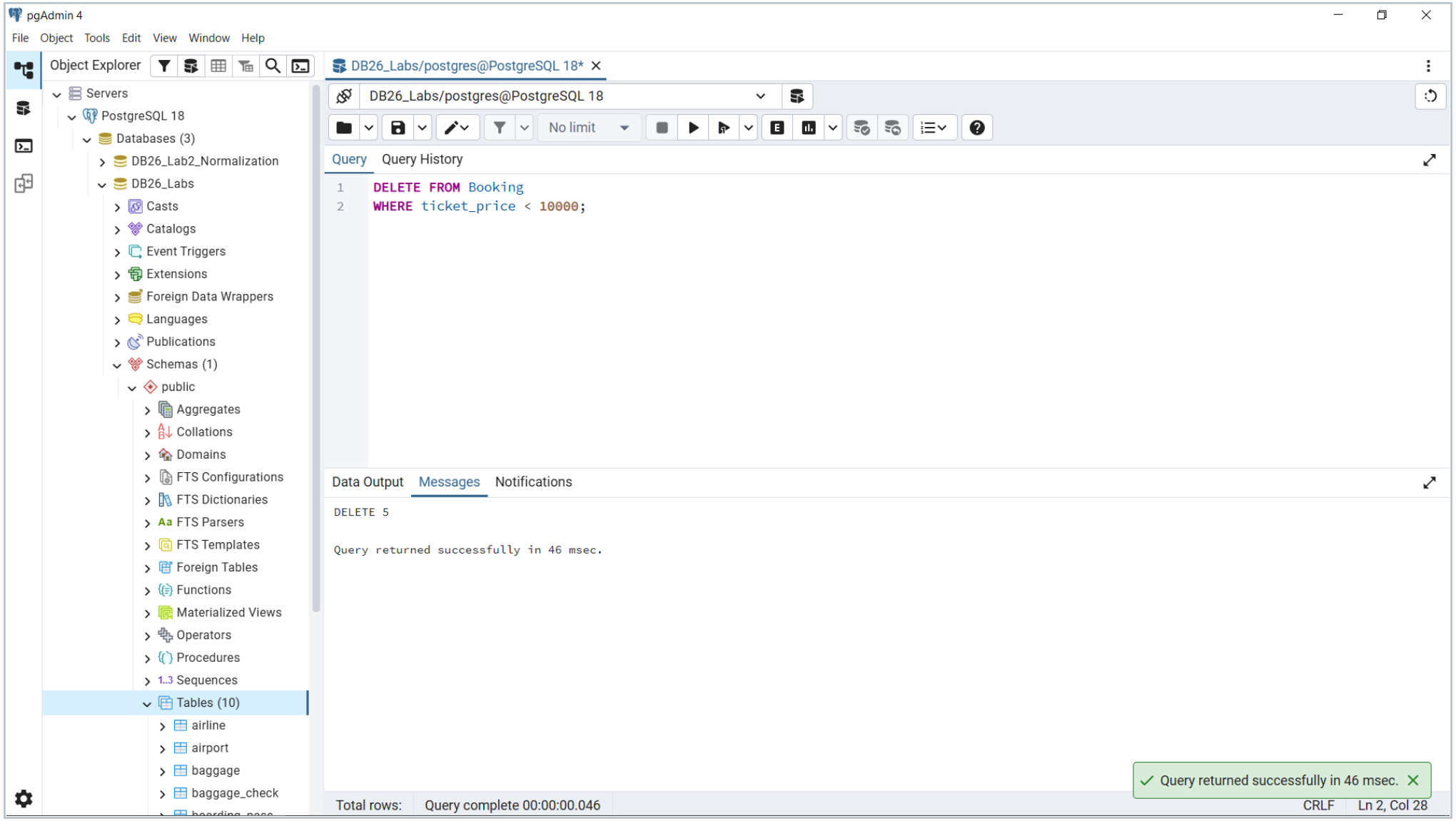
Description: Ticket price values after the update.



Execution Evidence 14

Screenshot_34.png

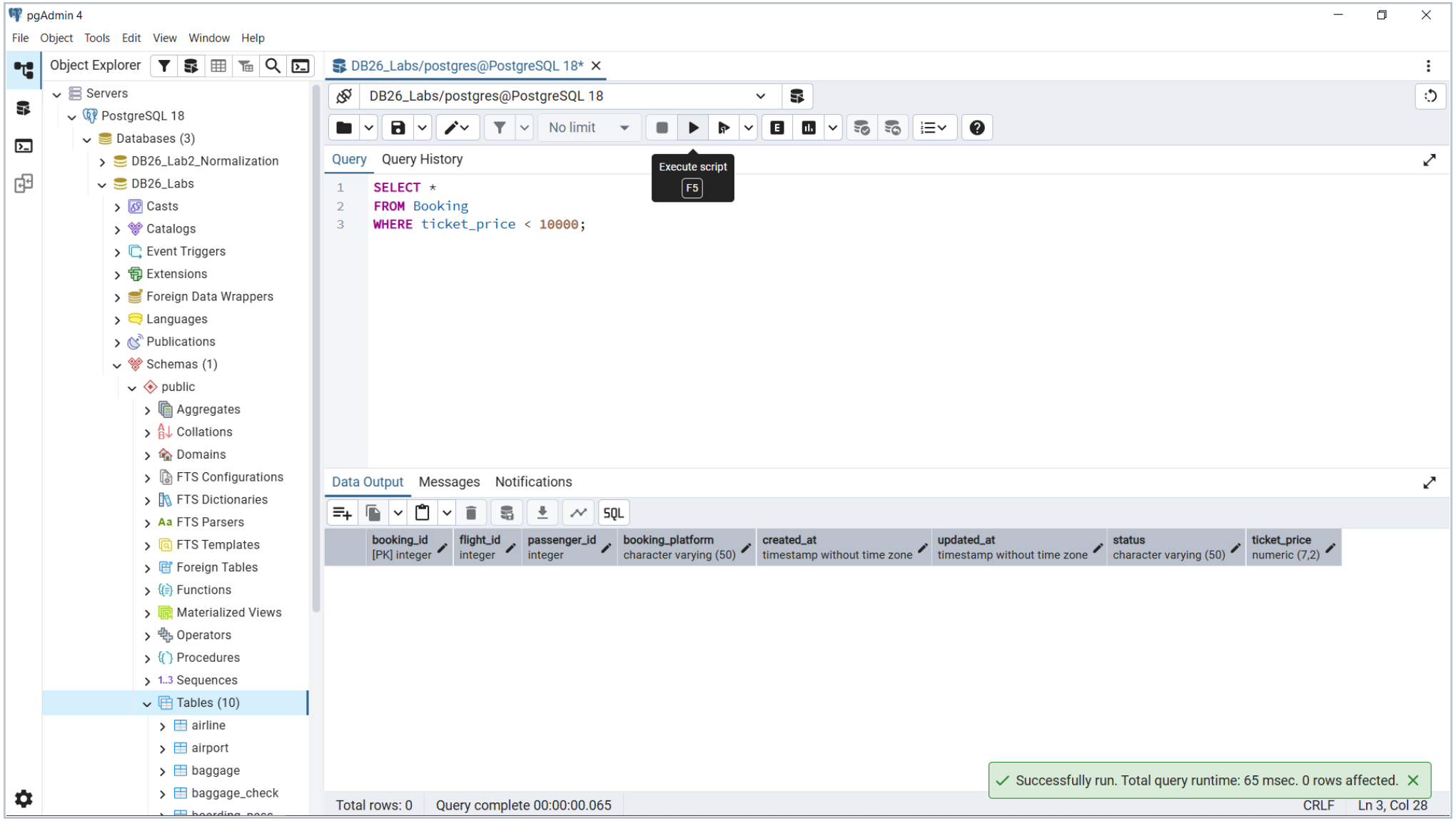
Description: DELETE operation removing bookings with ticket prices below the threshold.



Execution Evidence 15

Screenshot_35.png

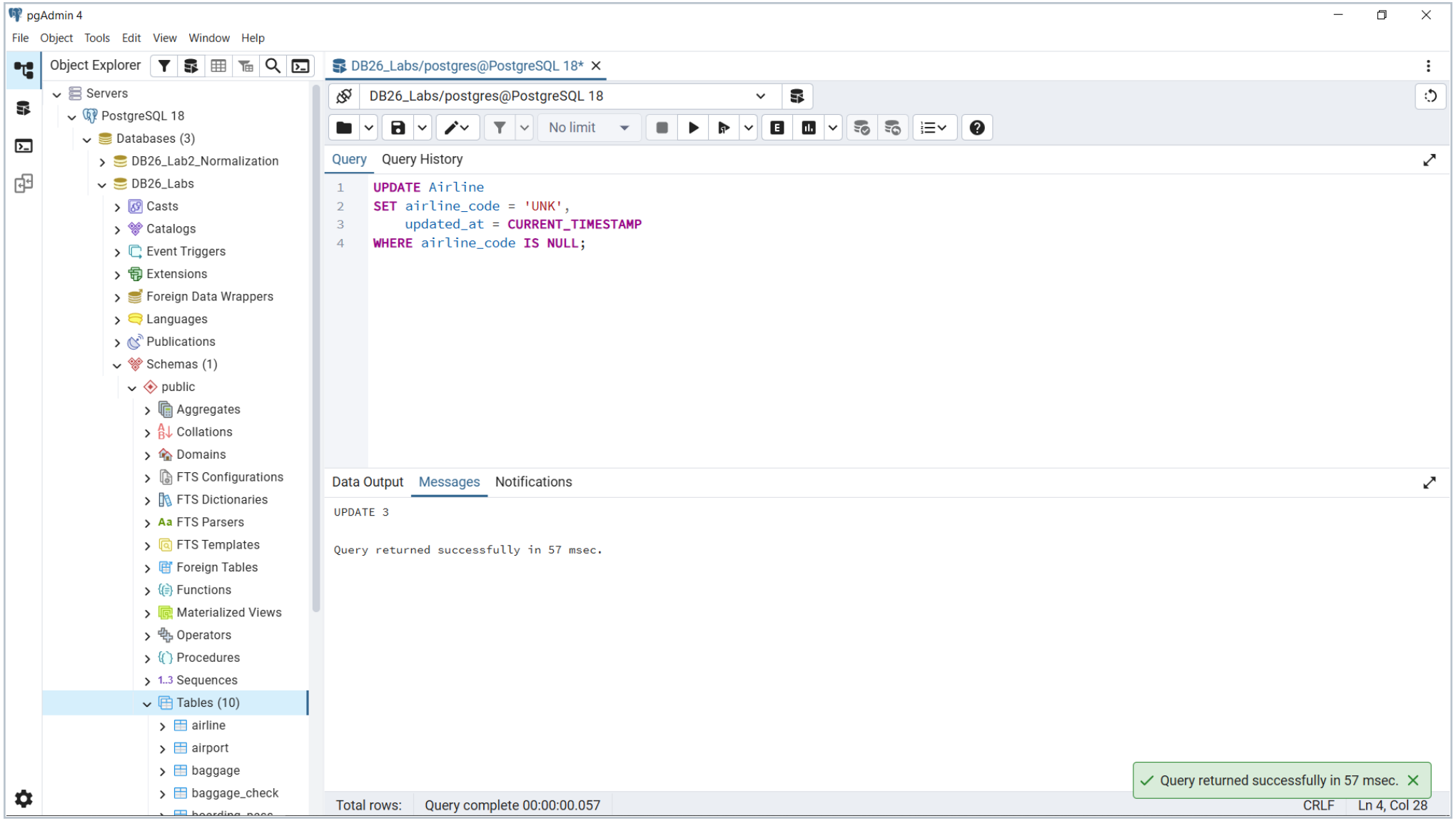
Description: Verification that no bookings below the threshold remain.



Execution Evidence 16

Screenshot_36.png

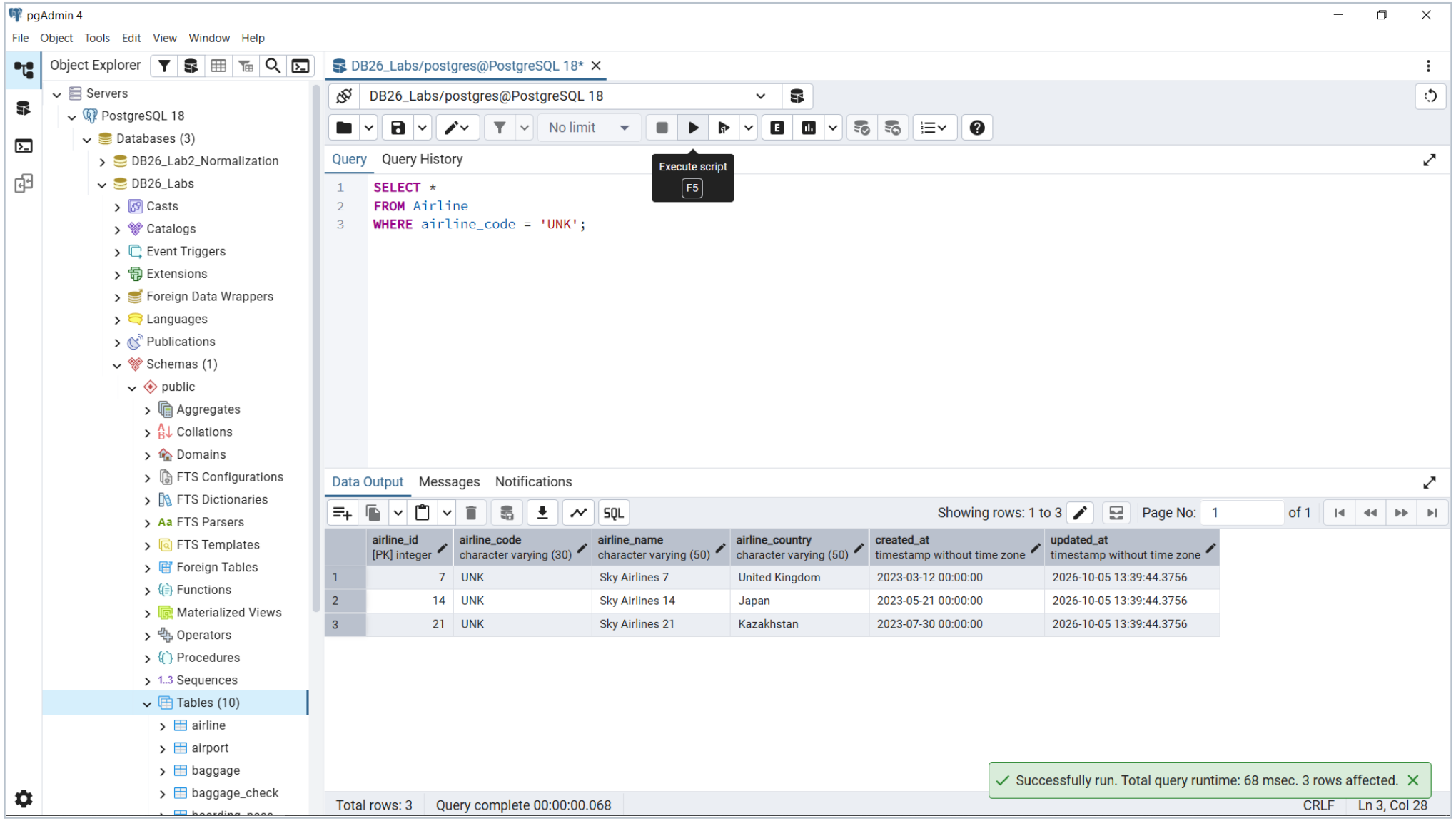
Description: UPDATE operation replacing NULL airline codes with 'UNK'.



Execution Evidence 17

Screenshot_37.png

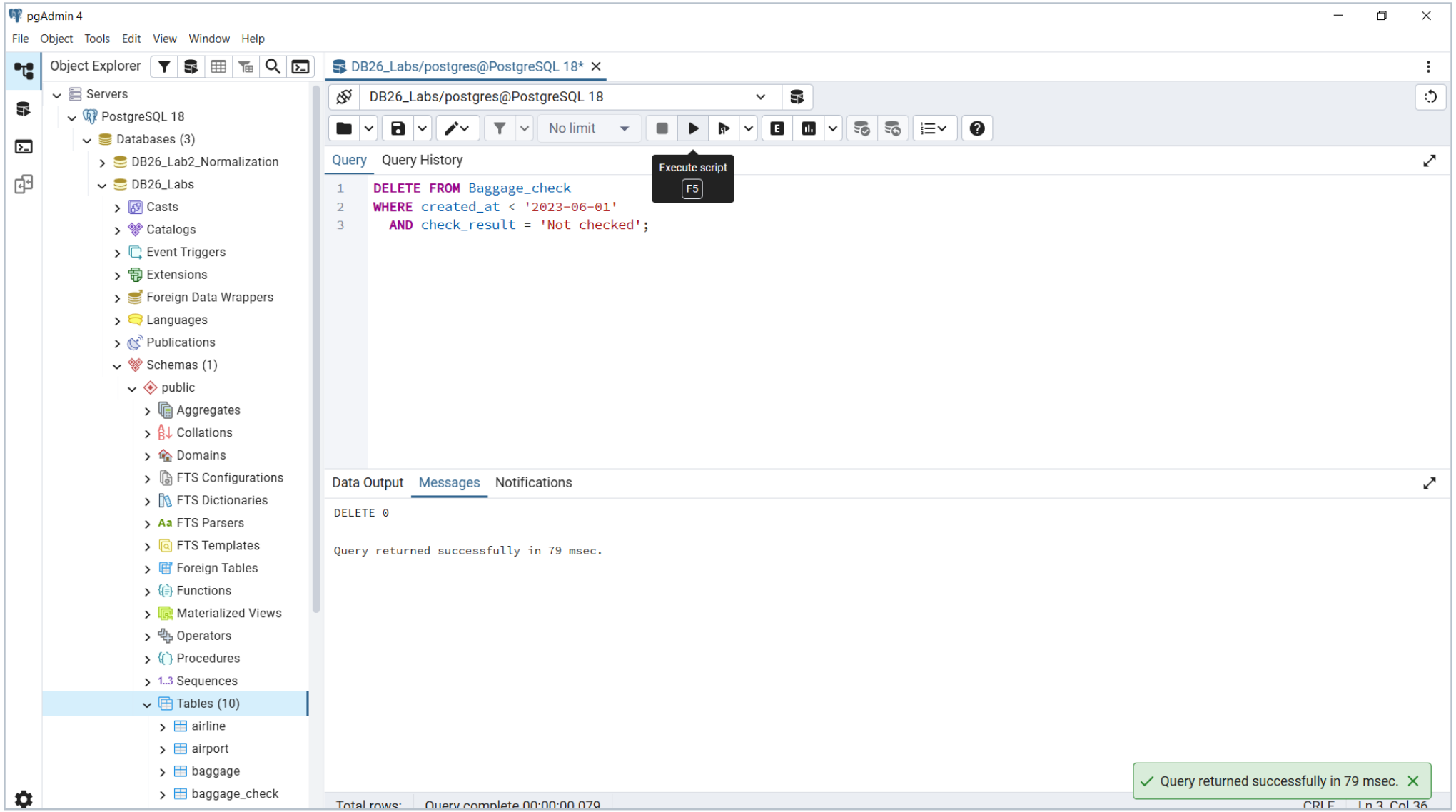
Description: Verification of airlines whose code was changed to 'UNK'.



Execution Evidence 18

Screenshot_38.png

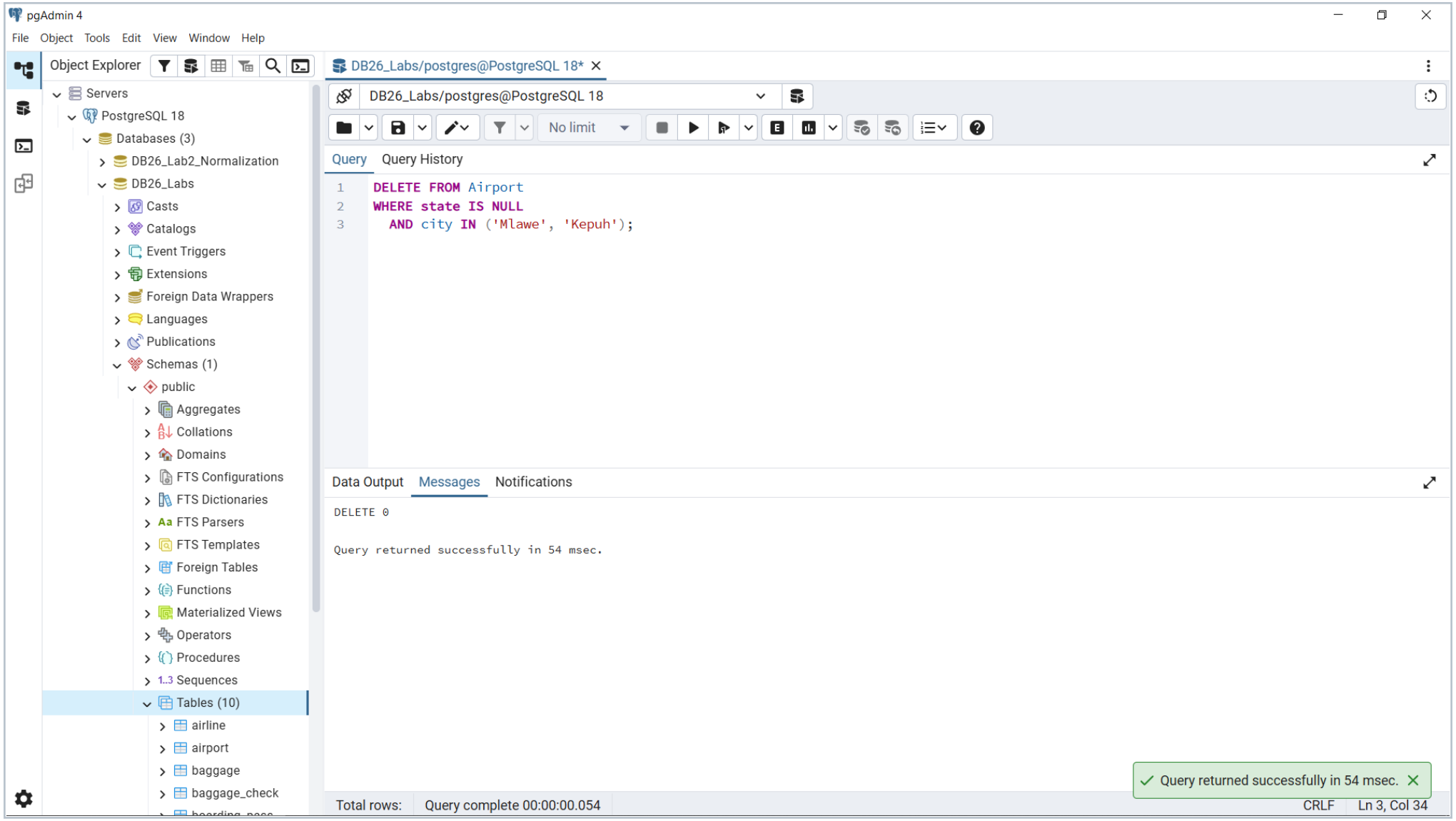
Description: Conditional DELETE operation on baggage check records.



Execution Evidence 19

Screenshot_39.png

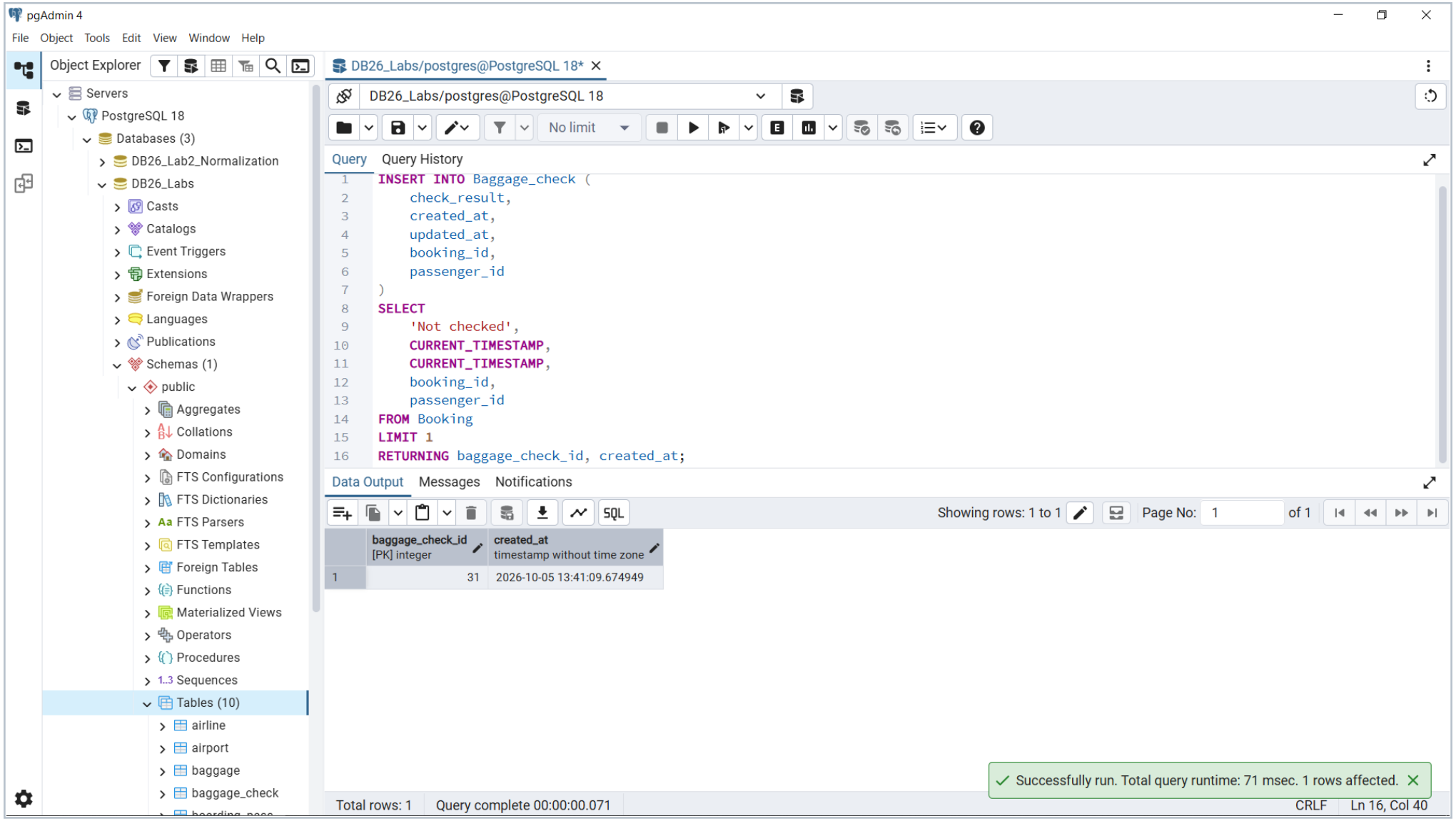
Description: Conditional DELETE operation on airport records.



Execution Evidence 20

Screenshot_40.png

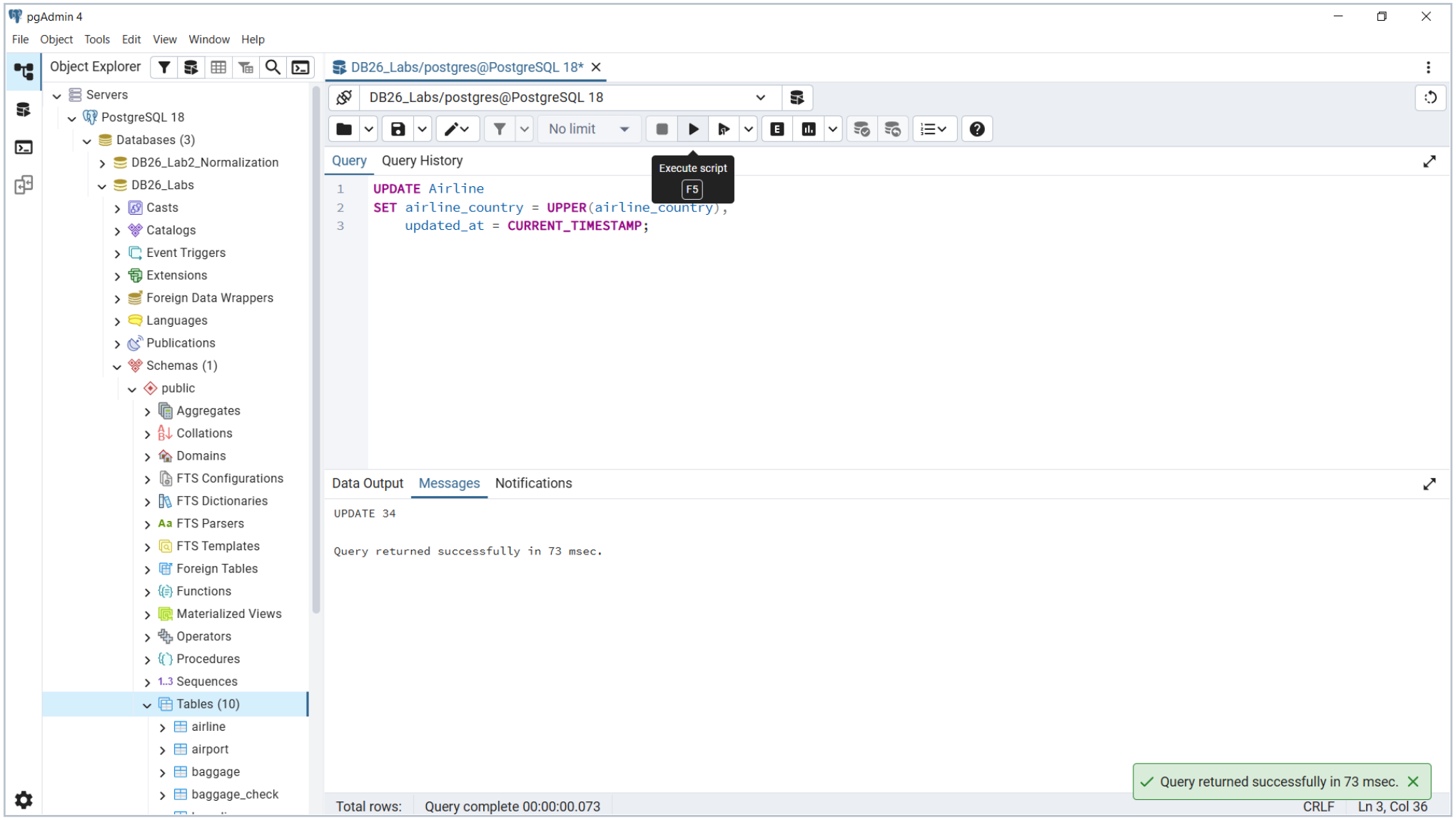
Description: Insertion of a baggage check record and returned identifier.



Execution Evidence 21

Screenshot_41.png

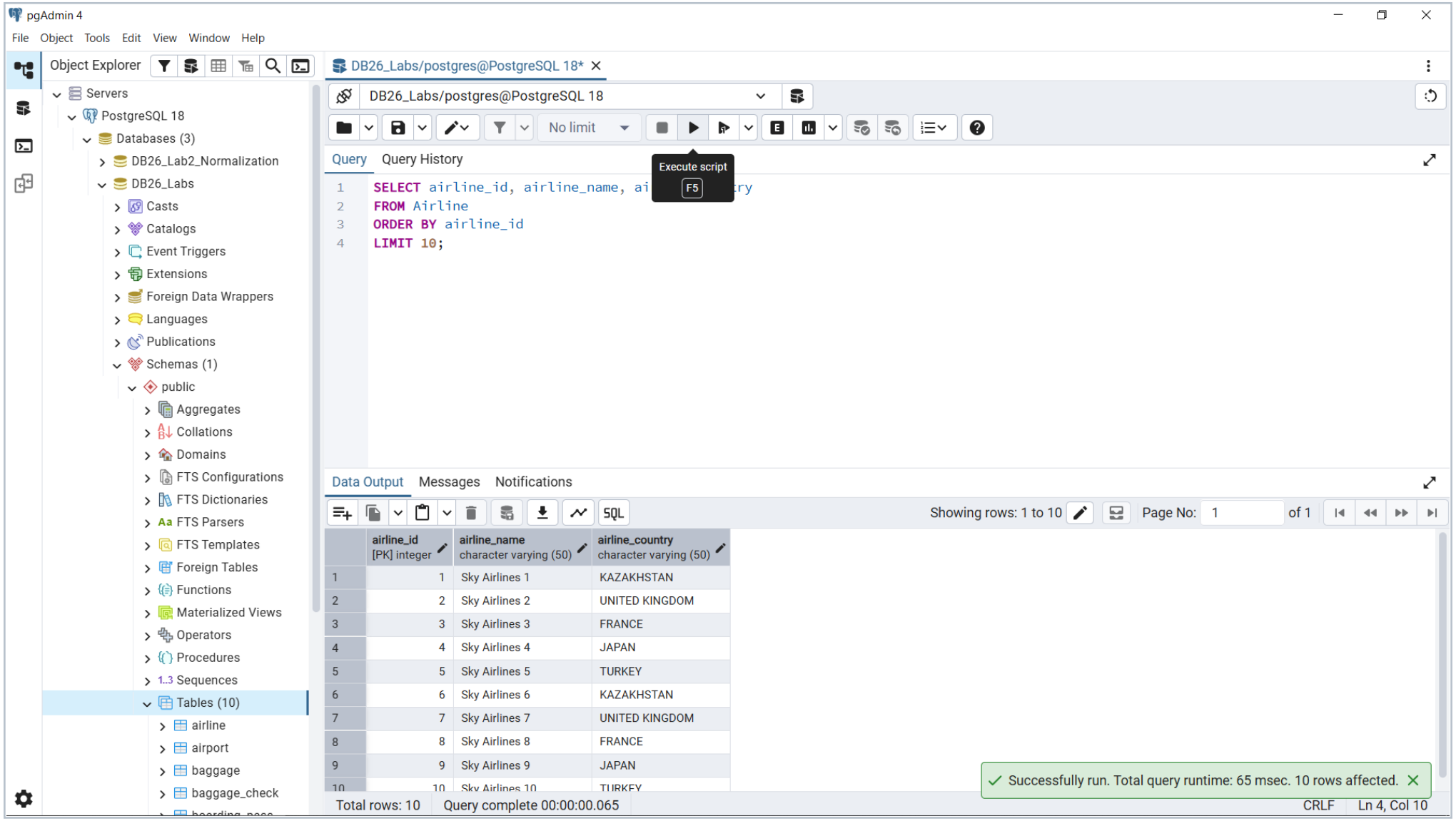
Description: Insertion of security check records.



Execution Evidence 22

Screenshot_42.png

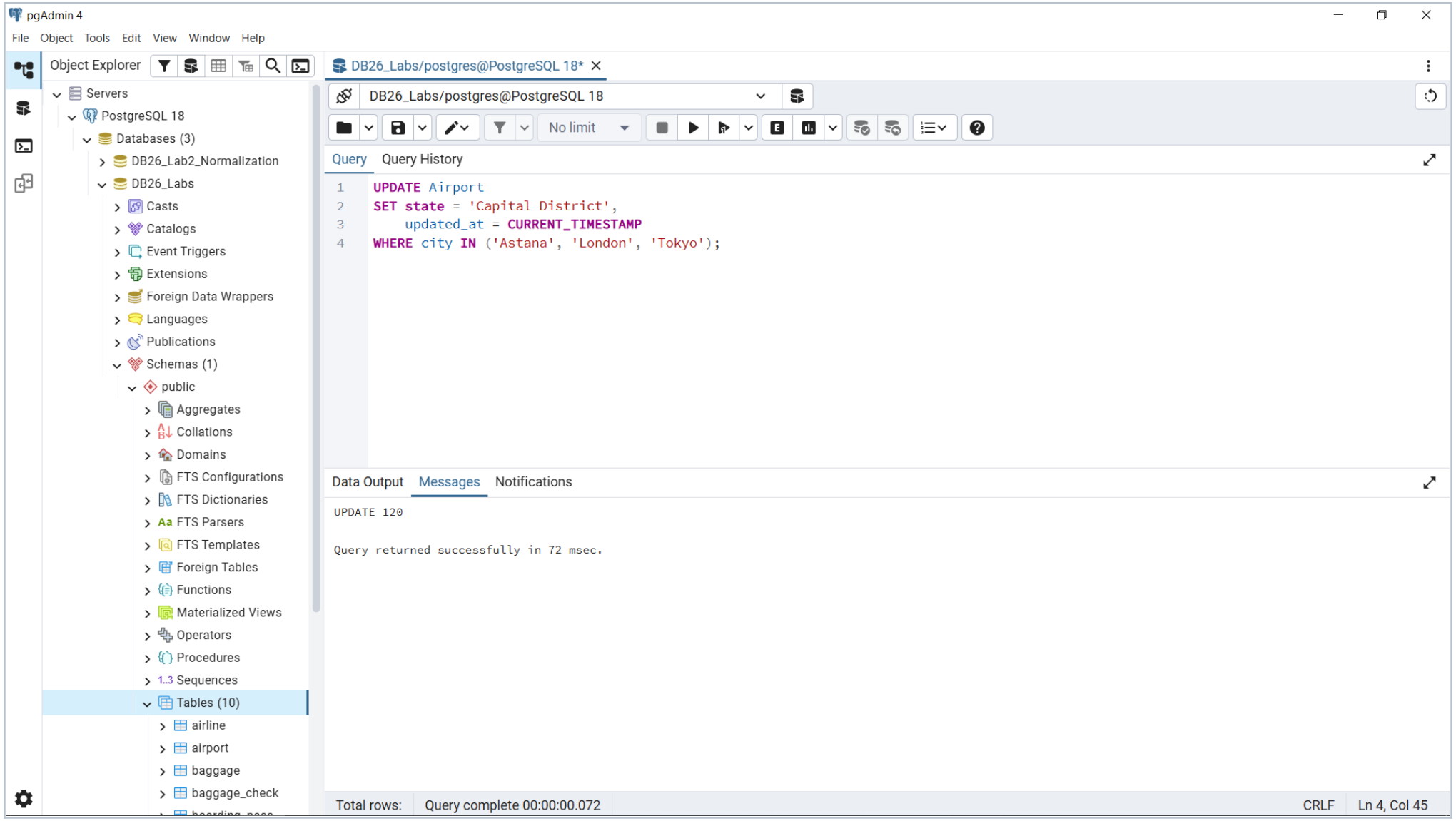
Description: Successful execution of the security check data population.



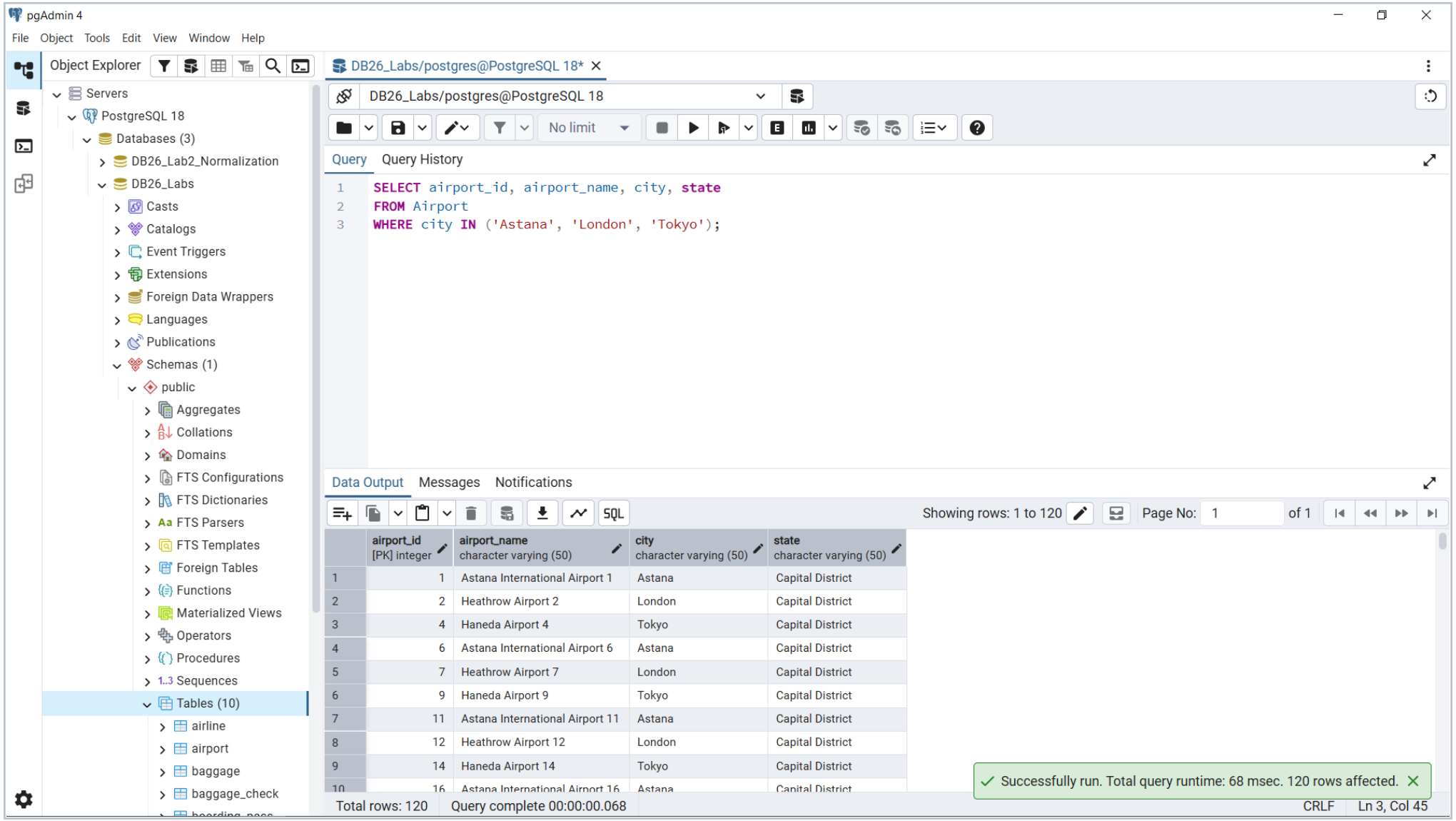
Execution Evidence 23

Screenshot_45.png

Description: UPDATE operation assigning 'Capital District' to selected airport cities.



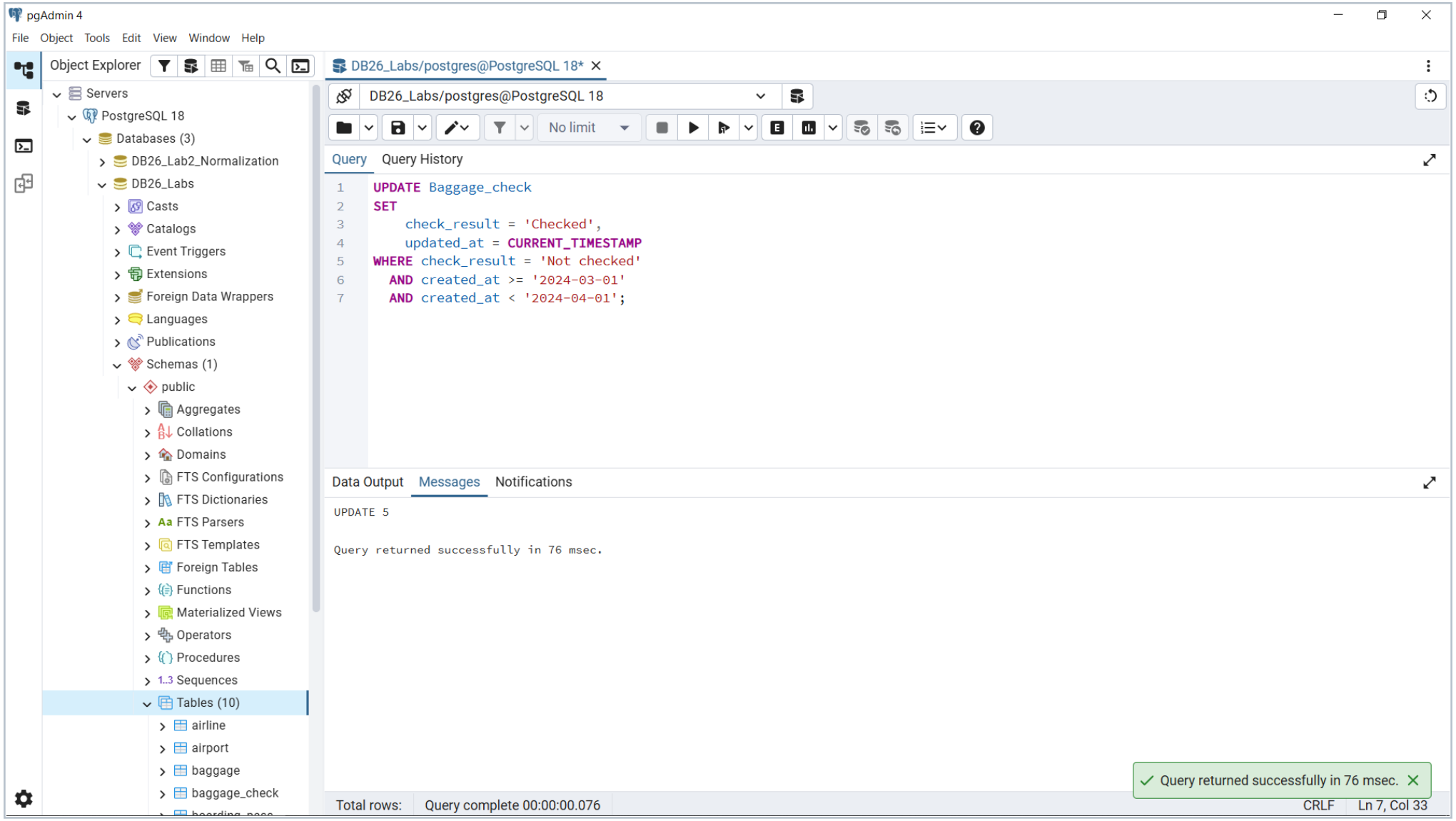
Description: Verification of airport state values after the update.



Execution Evidence 25

Screenshot_47.png

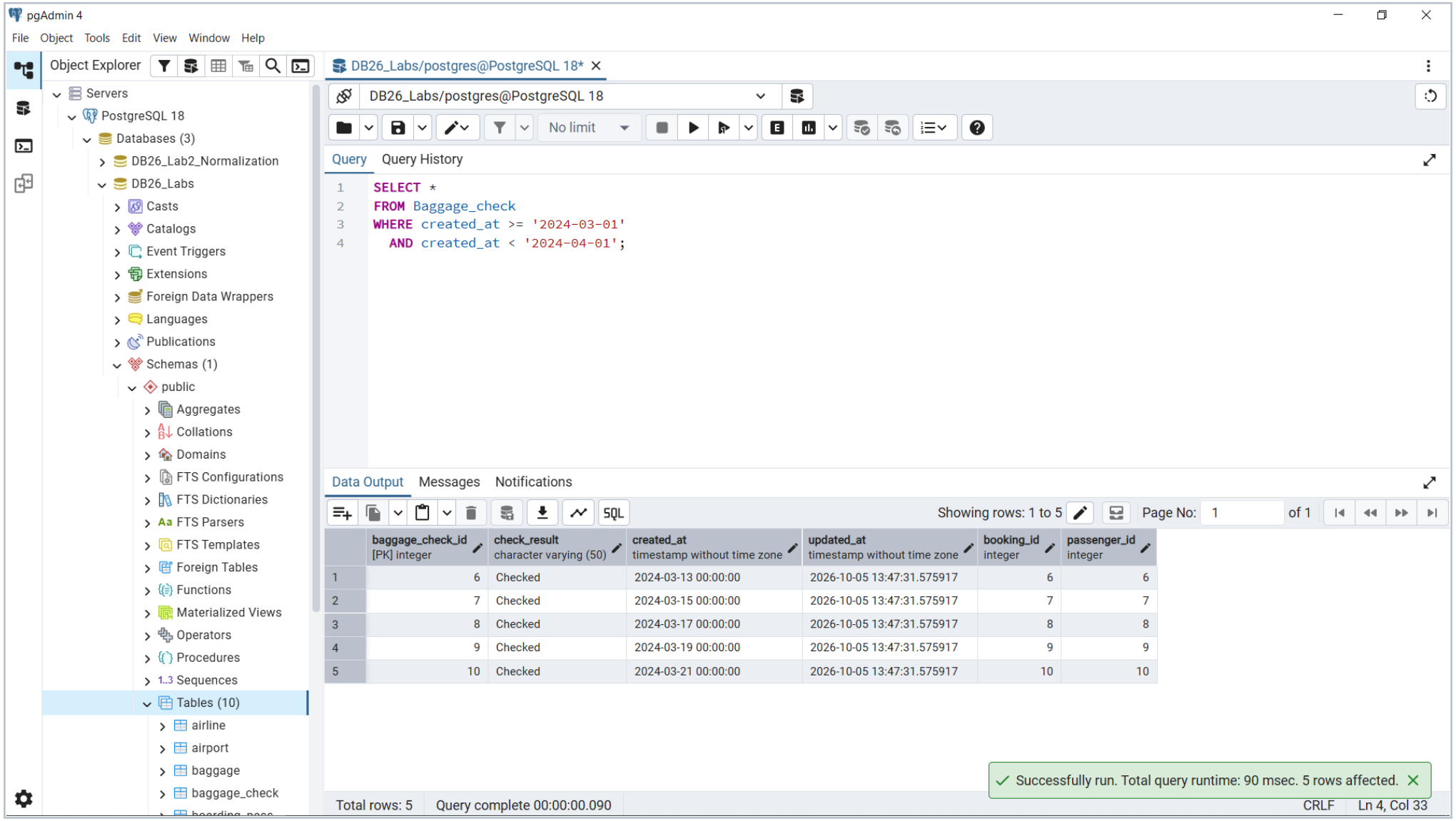
Description: UPDATE operation changing selected baggage check results to 'Checked'.



Execution Evidence 26

Screenshot_48.png

Description: Verification query showing the updated baggage check records.



Conclusion

During this laboratory work, the database was populated with test data and the main SQL data manipulation commands were applied. INSERT statements were used to create sample records, UPDATE statements changed selected values, DELETE statements removed records matching specified conditions, and SELECT queries were used to confirm the results. The screenshots demonstrate successful query execution in PostgreSQL through pgAdmin.